

FAST — National University of Computer and Emerging Sciences
Islamabad Campus ♦ Department of Computer Science

CS — Deep Learning | PROJECT PROPOSAL

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**SAM-Enhanced Spatially-Aware Multiple Instance Learning
for Whole Slide Image Classification**

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1. Abstract

This proposal presents a deep learning project targeting Whole Slide Image (WSI) classification for histopathology — one of the most computationally demanding and clinically critical tasks in computational pathology. We build directly upon the DTFD-MIL (Double-Tier Feature Distillation Multiple Instance Learning) framework presented at CVPR 2022, which introduced pseudo-bag augmentation and a two-tier attention-based MIL architecture to address small sample cohort limitations inherent in WSI analysis.

Our key innovation is the integration of the Segment Anything Model (SAM) to provide spatially-aware context during both pseudo-bag construction and the attention-based feature distillation process. Rather than randomly splitting patches into pseudo-bags, we propose using SAM-generated tissue segmentation masks to construct semantically coherent pseudo-bags, ensuring that spatially proximate, biologically related patches are grouped together. Additionally, we investigate replacing the ResNet-50 feature extractor with PLIP (Pathology Language-Image Pretraining) embeddings trained on 208,414 pathology images, to evaluate whether domain-specific visual-language features improve MIL classification performance.

Target outcomes include AUC improvement from 0.946 to 0.960+ on CAMELYON-16 and from 0.961 to 0.970+ on TCGA Lung Cancer, with qualitatively improved spatial localization of tumor regions compared to standard attention heatmaps. The complete implementation will be made publicly available on GitHub, validated on NVIDIA RTX-class hardware accessible to all team members.

2. Introduction and Background

2.1. Whole Slide Image Classification and the MIL Paradigm

Histopathology Whole Slide Images (WSIs) are extraordinarily large digital scans of tissue specimens, ranging from 100 million to 10 billion pixels in resolution. This scale makes direct application of standard deep learning classification architectures infeasible, as such models were designed for images orders of magnitude smaller. Instead, WSIs are divided into hundreds or thousands of non-overlapping patches, and classification is performed at the bag (slide) level — a paradigm formally known as Multiple Instance Learning (MIL).

Under MIL, a WSI (bag) contains many patches (instances), and the slide is labeled positive if at least one patch contains pathological evidence of disease. The challenge is that only slide-level labels are available during training, not patch-level annotations. This weakly supervised setting severely limits what information can guide model optimization, and in practice, histopathology cohorts contain only a few hundred to a few thousand labeled slides — creating significant risk of overfitting.

2.2. Baseline Framework: DTFD-MIL (CVPR 2022)

Our primary baseline is the Double-Tier Feature Distillation MIL (DTFD-MIL) framework by Zhang et al. (CVPR 2022). DTFD-MIL addresses the small-cohort problem by introducing the concept of pseudo-bags: the patches of a slide are randomly split into M smaller pseudo-bags, each inheriting the parent slide's label. This virtually multiplies the number of training bags, significantly reducing overfitting.

The framework operates on two tiers: a Tier-1 AB-MIL model is applied independently to each pseudo-bag, and the resulting features are distilled into compact representations that are then processed by a Tier-2 AB-MIL model to produce the final slide-level prediction. A key innovation in DTFD-MIL is the derivation of per-instance (per-patch) probability scores using a Grad-CAM inspired mechanism applied to the attention-based architecture — enabling more principled feature distillation than simple attention score selection.

DTFD-MIL achieves state-of-the-art AUC of 0.946 on CAMELYON-16 and 0.961 on TCGA Lung Cancer with the AFS and MaxMinS distillation strategies, respectively, surpassing RNN-MIL, Classic AB-MIL, DS-MIL, CLAM, and Trans-MIL. However, its pseudo-bag construction is purely random, ignoring the rich spatial relationships between patches in a WSI.

2.3. The Segment Anything Model (SAM) in Medical Imaging

The Segment Anything Model (SAM), introduced by Meta AI (Kirillov et al., 2023), is a promptable segmentation foundation model trained on over 1 billion masks across 11 million images. SAM demonstrates remarkable zero-shot and few-shot generalization to novel image domains including medical images. A growing body of literature (as surveyed in 'SAM for Medical Image Segmentation: Current Applications and Future Directions', Computational and Structural Biotechnology Journal, 2024) demonstrates SAM's applicability to pathology, radiology, and microscopy imaging.

Critically for our task, SAM can be applied to individual WSI patches to generate tissue-structure-aware segmentation masks — separating tumor regions, stroma, necrotic tissue, and background within each patch. These masks can serve as spatial priors that guide how patches are grouped into pseudo-bags, ensuring that semantically coherent tissue regions are considered together rather than arbitrarily split.

2.4. PLIP: Pathology-Specific Visual-Language Embeddings

Huang et al. (Nature Medicine, 2023) introduced PLIP (Pathology Language-Image Pretraining), a multimodal AI model trained on OpenPath — a dataset of 208,414 pathology images paired with natural language descriptions harvested from medical Twitter. PLIP achieves state-of-the-art performance on zero-shot pathology image classification, significantly outperforming general-domain CLIP models (zero-shot F1: 0.565–0.832 vs. 0.030–0.481 for prior models).

PLIP's domain-specific pretraining on actual pathology images and clinical descriptions makes it a promising alternative to ImageNet-pretrained ResNet-50 as the patch feature extractor in MIL pipelines. We hypothesize that PLIP's embeddings encode richer histopathological semantics, which will improve the quality of features fed into both tiers of the DTFD-MIL framework.

3. Problem Statement

Despite DTFD-MIL's state-of-the-art performance, two fundamental limitations remain unaddressed:

First, pseudo-bag construction is spatially blind. Patches from biologically contiguous tissue regions — such as a continuous tumor margin — may be split across different pseudo-bags purely by chance. This means the Tier-1 MIL model may receive pseudo-bags with incoherent spatial context, weakening the quality of distilled features passed to Tier-2. In histopathology, spatial tissue architecture is a critical diagnostic cue: the arrangement and boundary characteristics of tumor cells relative to surrounding stroma are often more informative than individual patches in isolation.

Second, ResNet-50 pretrained on ImageNet carries a domain gap when applied to stained histopathology slides. ImageNet features encode natural scene statistics that differ substantially from the morphological, staining, and texture patterns diagnostic in pathology. While PLIP demonstrates that pathology-specific pretraining dramatically improves classification performance in zero-shot settings, its benefit within a MIL pipeline — particularly in interaction with pseudo-bag distillation — has not been studied.

Our research questions are therefore:

1. Can SAM-guided spatial pseudo-bag construction — grouping patches by tissue structure rather than randomly — improve WSI classification AUC on CAMELYON-16 and TCGA Lung Cancer?
2. Does incorporating SAM-derived spatial mask embeddings into the attention mechanism provide additional localization accuracy for tumor region identification?
3. Do PLIP embeddings outperform ResNet-50 as the patch feature extractor within the DTFD-MIL framework, and do they interact positively with SAM-guided pseudo-bags?

4. Motivation

4.1. Clinical Motivation

Accurate and interpretable WSI classification directly supports pathologists in diagnosing cancer. CAMELYON-16 addresses lymph node metastasis detection in breast cancer — a critical staging decision that determines treatment protocol. Improvement in AUC from 0.946 to 0.960+ translates directly to fewer missed metastases (false negatives) and fewer unnecessary further investigations (false positives). Improved spatial localization via SAM-enhanced heatmaps would provide pathologists with more reliable visual guidance, moving toward clinically trustworthy AI assistance.

4.2. Scientific Motivation

The integration of foundation models (SAM) with domain-specific pretraining (PLIP) and specialized MIL architectures (DTFD-MIL) represents a convergence of the most powerful recent advances in vision AI for pathology. Each component has been separately validated, but their combined interaction within a unified framework is unexplored. This project contributes a principled study of how spatial priors from SAM can bridge the gap between patch-level feature quality and slide-level prediction accuracy — a fundamental question for the broader MIL research community.

4.3. Educational Motivation

For our team at FAST NUCES, this project provides comprehensive exposure to: attention-based MIL architectures and weakly supervised learning; foundation model fine-tuning and zero-shot transfer (SAM, PLIP); WSI preprocessing pipelines (OpenSlide, OTSU thresholding, patch extraction); ablation study design for isolating contributions of architectural components; and the full research paper writing process from methodology through experimental evaluation.

5. Project Objectives

No.	Objective	Baseline	Target
O1	Integrate SAM-generated spatial context masks into DTFD-MIL pseudo-bag construction	Random spatial split	SAM-guided spatial split
O2	Achieve AUC improvement on CAMELYON-16 over DTFD-MIL baseline	AUC 0.946 (AFS)	AUC ≥ 0.960
O3	Achieve AUC improvement on TCGA Lung Cancer over baseline	AUC 0.961 (MaxMinS)	AUC ≥ 0.970
O4	Design a spatially-aware attention mechanism using SAM region embeddings	Standard AB-MIL attention	Spatial-context attention
O5	Validate PLIP embeddings as feature extractor vs. ResNet-50 baseline	ResNet-50 features	PLIP + ResNet comparison
O6	Reproduce DTFD-MIL baseline on both datasets	Published results	Reproduced within $\pm 1\%$

6. Technical Contributions

6.1. Contribution 1: SAM-Guided Spatial Pseudo-Bag Construction

The central innovation of this project is replacing DTFD-MIL's random pseudo-bag splitting with a spatially-aware construction strategy. For each WSI, after patch extraction:

4. SAM is run on each 256x256 patch to generate fine-grained tissue segmentation masks, identifying tumor, stroma, necrosis, and background regions.
5. Patch-level spatial context vectors are computed by aggregating mask statistics (region areas, boundary lengths, region counts) into a fixed-size descriptor.
6. Patches from the same slide are then clustered into M pseudo-bags using K-means on these spatial context vectors, ensuring each pseudo-bag contains a spatially coherent subset of tissue types rather than a random mixture.

This approach guarantees that each Tier-1 pseudo-bag processes a semantically consistent micro-environment, improving the quality of distilled features sent to Tier-2. Importantly, the label noise problem inherent in random splitting — where a pseudo-bag from a positive slide may contain zero positive patches — is partially mitigated because spatial clustering tends to group patches by their tissue type, concentrating positive (tumor) patches into a smaller number of pseudo-bags.

6.2. Contribution 2: SAM-Enhanced Spatial Attention

Beyond pseudo-bag construction, we augment the attention mechanism itself with SAM-derived spatial information. In standard AB-MIL, the attention score for patch k is computed solely from its feature vector h_k . We extend this by concatenating SAM spatial context descriptors with patch features before the attention module, enabling the network to explicitly reason about where in the tissue architecture each patch lies when computing its importance weight.

This spatial attention augmentation is inspired by the observation from DTFD-MIL's Figure 4: standard attention scores produce strong false-positive activations in negative slides, highlighting neutral or blank regions. SAM

context features can help the network suppress attention over tissue-absent or morphologically unremarkable regions, reducing false positive heatmap activations.

6.3. Contribution 3: PLIP Feature Extractor Integration

We replace the ResNet-50 (ImageNet-pretrained) patch feature extractor with PLIP image embeddings (512-dimensional), which were trained specifically on pathology images and their clinical descriptions. PLIP is used in image-only mode (without text prompts) as a fixed feature extractor, with an optional learnable linear projection to match the feature dimensionality expected by the AB-MIL attention modules.

We conduct a controlled comparison: (a) ResNet-50 + standard pseudo-bags (DTFD-MIL baseline), (b) PLIP + standard pseudo-bags, (c) ResNet-50 + SAM pseudo-bags, and (d) PLIP + SAM pseudo-bags — allowing us to isolate and quantify each contribution independently.

6.4. Contribution 4: Spatial Localization Analysis

Extending DTFD-MIL's instance probability derivation (Equations 8–9), we generate improved tumor localization heatmaps that combine Grad-CAM derived instance probabilities with SAM mask overlay. The resulting visualization shows not only which patches have high tumor probability, but also which tissue structures within each patch are most diagnostically relevant — providing pathologists with more interpretable and localized evidence.

7. Methodology

7.1. Phase 1: Baseline Reproduction and Profiling

7. Obtain and preprocess CAMELYON-16 and TCGA Lung Cancer datasets; apply OTSU thresholding to localize tissue regions; extract 256x256 patches at 20x magnification.
8. Implement and train DTFD-MIL with all four feature distillation strategies (MaxS, MaxMinS, AFS, MAS) using ResNet-50 features; reproduce published AUC scores within $\pm 1\%$.
9. Profile training time, memory consumption, and AUC curves; establish evaluation harness for systematic comparison.

7.2. Phase 2: SAM Integration

10. Run SAM (or MedSAM for pathology-specific performance) on all extracted patches; store segmentation masks as binary arrays alongside patch features.
11. Compute spatial context descriptors per patch: (a) number of distinct tissue segments, (b) area fractions per segment class, (c) segment boundary density, (d) centroid distribution.
12. Implement K-means based pseudo-bag splitter using spatial descriptors; validate that spatial clustering produces more homogeneous pseudo-bags than random splitting using a homogeneity score on tissue-type labels from a subset of annotated slides.

7.3. Phase 3: Model Development

13. Integrate PLIP feature extractor: load pretrained PLIP image encoder, add projection layer ($512 \rightarrow 512$), freeze encoder weights, train projection + AB-MIL modules end-to-end.
14. Implement spatial attention augmentation: concatenate SAM descriptors (d-dimensional) to patch features before the attention network; ensure no gradient flows back into SAM (frozen).
15. Train full SAM-PLIP-DTFD-MIL model with cross-entropy loss on both tiers (Equations 11 + 13) using AdamW optimizer, cosine LR scheduler, and early stopping on validation AUC.

7.4. Phase 4: Ablation Studies

We conduct systematic ablation across five dimensions to isolate each contribution:

- Feature extractor: ResNet-50 vs. PLIP
- Pseudo-bag construction: random vs. SAM-guided
- Attention augmentation: with vs. without SAM spatial descriptors
- Number of pseudo-bags M : sweep from 2 to 10
- Distillation strategy: MaxS vs. MaxMinS vs. AFS vs. MAS

7.5. Phase 5: Evaluation and Reporting

16. Evaluate all configurations on CAMELYON-16 test set (5-run mean $\pm$ CI-95) and TCGA Lung Cancer (4-fold cross-validation).
17. Generate SAM-enhanced localization heatmaps for 5 representative slides; compare visually against DTFD-MIL attention heatmaps and against published ground-truth tumor outlines.
18. Produce IEEE-format final report documenting methodology, results, ablation tables, and heatmap visualizations.

8. Datasets

8.1. CAMELYON-16

Type: Lymph node metastasis detection in breast cancer WSIs.

- Training: 270 slides (160 normal, 110 tumor)
- Test: 130 slides — official train/val split at 9:1 ratio
- Patch size: 256x256 pixels at 20x magnification; ~3.7 million patches total
- Challenge: Positive tumor regions occupy only small portions of positive slides — requiring models that correctly identify rare positive instances

This is the primary evaluation dataset as it represents the hardest case for MIL (small positive regions), making it the most sensitive testbed for spatial context improvements.

8.2. TCGA Lung Cancer

Type: Lung cancer subtype classification (LUAD vs. LUSC) from The Cancer Genome Atlas.

- Split: 65% train / 10% validation / 25% test (patient-level); 4-fold cross-validation
- ~8.3 million patches total across all slides
- Tumor occupies larger proportions of positive slides — lower difficulty than CAMELYON-16 but provides complementary evaluation

9. Baseline Paper and Related Work

9.1. Primary Baseline

Zhang H., Meng Y., Zhao Y., Qiao Y., Yang X., Coupland S.E., Zheng Y. 'DTFD-MIL: Double-Tier Feature Distillation Multiple Instance Learning for Histopathology Whole Slide Image Classification.' CVPR 2022. <https://github.com/hrzhang1123/DTFD-MIL>

9.2. Related Technical Literature

- 19. Kirillov A. et al. 'Segment Anything.' ICCV 2023 — foundation model providing spatial segmentation for WSI patches.
- 20. Huang Z. et al. 'A visual-language foundation model for pathology image analysis using medical Twitter.' Nature Medicine, 2023 — PLIP pathology embeddings trained on OpenPath (208,414 images).
- 21. 'SAM for Medical Image Segmentation: Current Applications and Future Directions.' Computational and Structural Biotechnology Journal, 2024 — survey of SAM in clinical imaging.
- 22. Ilse M., Tomczak J., Welling M. 'Attention-based Deep Multiple Instance Learning.' ICML 2018 — Classic AB-MIL, foundational architecture for all tiers in our model.
- 23. Lu M.Y. et al. 'Data-efficient and weakly supervised computational pathology on whole-slide images.' Nature Biomedical Engineering, 2021 — CLAM baseline.
- 24. Shao Z. et al. 'TransMIL: Transformer Based Correlated Multiple Instance Learning for Whole Slide Image Classification.' NeurIPS 2021 — transformer-based MIL baseline.
- 25. Li B., Li Y., Eliceiri K.W. 'Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning.' CVPR 2021 — DS-MIL baseline.
- 26. Selvaraju R.R. et al. 'Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization.' ICCV 2017 — mechanism underlying instance probability derivation in DTFD-MIL.

10. Expected Outcomes

10.1. Performance Targets

Metric	Baseline	Expected Target
AUC (CAMELYON-16)	0.946 (DTFD-MIL AFS)	≥ 0.960
AUC (TCGA Lung Cancer)	0.961 (DTFD-MIL MaxMinS)	≥ 0.970
F1 Score (CAMELYON-16)	0.882 (DTFD-MIL AFS)	≥ 0.900
Spatial Localization Quality	Attention score heatmap	SAM-guided + derived probability heatmap
Feature Extractor	ResNet-50 (ImageNet)	PLIP + ResNet-50 comparison

10.2. Project Deliverables

- 27. Complete PyTorch implementation of SAM-Enhanced DTFD-MIL, committed to a public GitHub repository with documented training scripts.
- 28. Precomputed SAM mask dataset for CAMELYON-16 and TCGA Lung Cancer patches (stored as .npy arrays, made publicly available).
- 29. IEEE double-column format final report (10 pages) covering methodology, ablation study results, localization heatmaps, and statistical comparisons.
- 30. Ablation experiment logs exported from Weights & Biases covering all configurations.
- 31. Final presentation slide deck (15 minutes) with live heatmap visualization demo.

10.3. Scientific Impact

This project establishes the first systematic evaluation of SAM-guided pseudo-bag construction within a double-tier MIL framework. If successful, it will demonstrate that spatial priors from segmentation foundation models can meaningfully improve the quality of weakly supervised WSI classification — a finding with broad implications for computational pathology research. The integration of PLIP provides a parallel contribution:

establishing whether pathology-specific visual-language pretraining improves MIL performance beyond ImageNet baselines.

11. Project Timeline

Phase	Timeline	Key Activities
Phase 1: Baseline Reproduction	Weeks 1–2	Reproduce DTFD-MIL on CAMELYON-16 & TCGA; verify published AUC scores
Phase 2: SAM Integration	Weeks 3–4	Run SAM on patch datasets; generate spatial masks; design SAM-guided pseudo-bag splitter
Phase 3: Model Development	Weeks 5–7	Implement spatially-aware attention; integrate PLIP feature extractor; train combined model
Phase 4: Ablation Studies	Weeks 8–9	Isolate contributions of each component; test distillation strategies
Phase 5: Evaluation & Analysis	Weeks 10–11	Full benchmark on both datasets; comparison with DTFD-MIL, Trans-MIL, CLAM
Phase 6: Report & Presentation	Week 12	IEEE-format paper; final presentation slides

12. Hardware and Software Requirements

Component	Specification / Tool	Purpose
GPU	NVIDIA RTX 3080 / RTX 4060 (student laptops)	SAM inference + MIL model training
CPU	Intel Core i7 / AMD Ryzen 7 (16-core)	Data preprocessing & patch extraction
RAM	32 GB DDR4/DDR5	Patch caching & batch loading
Cloud GPU	Google Colab Pro / Kaggle (A100/T4)	Large-scale training runs
PyTorch 2.x	Deep learning framework	Model training & inference
Segment Anything Model	Meta AI (SAM / SAM-HQ / MedSAM)	Spatial mask generation for WSI patches
PLIP (OpenPath)	Pathology Language-Image Pretraining	Domain-specific feature extraction
OpenSlide	WSI reading library	Tile extraction from .svs/.tif files
OTSU Thresholding	Scikit-image	Tissue region detection in WSIs
Weights & Biases	Experiment tracking	AUC / F1 curve logging & ablation tracking

13. Risk Assessment and Mitigation

Risk	Likelihood	Mitigation Strategy
SAM inference cost on WSI patches	High	Use SAM-HQ or MedSAM with batch processing on GPU; precompute and cache all masks offline before training
PLIP embeddings incompatible with MIL pipeline dimensions	Medium	Implement projection head to align PLIP (512-d) to MIL feature space; fallback to ResNet-50
Overfitting on small CAMELYON-16 cohort	High	Pseudo-bag augmentation + dropout; early stopping on validation AUC
SAM masks noisy on stained histopathology	Medium	Fine-tune SAM decoder on 200 manually verified WSI patches; use MedSAM variant
Compute resource constraints	Low	Use Google Colab Pro / university GPU cluster; limit batch sizes

14. Team Responsibilities

Roll No.	Name	Primary Responsibilities
231-2598	Ahmed Baloch	Phase 1 baseline reproduction Phase 2 SAM mask generation & pseudo-bag integration, Phase 4 ablation studies, paper Sections 1–3
231-2596	Issa Sultan	Phase 3 spatially-aware attention design and PLIP feature extractor integration, Phase 4 distillation strategy comparison, paper Section 4–6
231-2564	Haseeb Siddiqui	Phase 3 model training pipeline, Phase 5 evaluation AUC and comparative analysis, Phase 6 report & presentation, paper Sections 7–9

15. References

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41. Campanella G. et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*, 25(8):1301–1309, 2019.